

World-space hand comparison on HOI4D: every method on OUR detector boxes

All metrics in mm, lower is better. *Every* learned row uses our predictive detection boxes (detbox v3), the same 157-sequence held-out set, and a 30-frame WA window, scored with our global metrics. C-MPJPE = *absolute* camera-frame joint error; WA-MPJPE = trajectory error after per-window alignment; W-MPJPE = global trajectory error. FPS is inference throughput (higher is better). **TBD** = recompute on detbox v3 in progress.

Method	Protocol	C-MPJPE ↓	WA-MPJPE ↓	W-MPJPE ↓	FPS ↑
<i>Offline, metric (per-frame metric model + DROID-SLAM trajectory), our detbox v3</i>					
HaWoR ^b	offline + SLAM	87.7	40.1	53.4	2.47 ^f
HaMeR _{mt} + SLAM ^g	offline + SLAM	67.3	41.8	66.8	2.47 ^f
HaMeR _{ft} + SLAM ^g	offline + SLAM	37.4	36.8	53.6	2.47 ^f
<i>Weak-perspective (no metric depth) – only WA-MPJPE is metric-meaningful, our detbox v3</i>					
WiLoR + SLAM ^a	offline + SLAM	–	33.4	–	2.47 ^f
HaPTIC + SLAM ^c	offline + SLAM	–	35.7	–	2.47 ^f
Dyn-HaMR ^h	offline optim.	–	48.3	–	2.47 ^f
<i>Online / feedforward</i>					
Hand3R (reported) ^d	online	42.6	38.0	86.9	–
Ours (feedforward) ^e	online, no SLAM	36.2	33.2	37.3	6.59

^a WiLoR + SLAM on detbox v3: per-frame WiLoR cropped from our detbox v3, composed with the same DROID-SLAM trajectory as our HaWoR run. WiLoR is *weak-perspective* (dummy focal), so its absolute placement is not metric: composed C-MPJPE and W-MPJPE are ~ 1 m regardless of resolution (verified on native-HD, focal-corrected: GT-box C-MPJPE 1152; the pose itself is good, root-relative 51 mm) - the documented weak-perspective failure. Hence only the alignment-invariant WA-MPJPE (33.4) is metric-meaningful. Vanilla HaMeR + SLAM behaves identically (weak-perspective); the metric HaMeR + SLAM rows are the absolute-depth-tuned HaMeR_{mt/ft} above. ^b HaWoR on detbox v3: the full pipeline (DROID-SLAM + Metric3D + motion + infiller) run end-to-end on all 157 sequences with our boxes injected in place of its own detector, 0 sequence failures. On its *own* tuned detector it scores 49.7 / 34.7 / 42.3; on our boxes it degrades to 87.7 / 40.1 / 53.4. This is the fair, input-matched number: our detbox v3 is genuinely noisier than HaWoR's detector (recall ~ 0.78 , IoU-vs-GT ~ 0.5), and HaWoR's motion+infiller is tuned to its own box statistics, so under a common realistic detector our jitter-robust model holds up while HaWoR does not. ^c HaPTIC + SLAM on detbox v3: HaPTIC is weak-perspective, so its metric depth is not recoverable. We verified this deeply on the 157-seq run - absolute C-MPJPE is 157.7 with a *depth-dependent* error (C-abs/C-rr ratio median 4.4, spread 2.4-8.0), while root-relative C-MPJPE is a healthy 29.2: the pose is good, only metric placement fails (the classic weak-perspective mode, amplified at the 224px crop focal). HaPTIC does no camera tracking, so for a fair world-space number we compose its per-frame camera-space hand with the *same* DROID-SLAM trajectory as the WiLoR/HaMeR+SLAM rows (shared HaWoR trajectory, recovered SE(3) residual 0.000 mm), then score identically. Only the alignment-invariant WA-MPJPE (35.7) is metric-meaningful; C-MPJPE and W-MPJPE are omitted as weak-perspective. ^d Hand3R: reported paper numbers; it cannot be re-run on our boxes without standing up its pipeline, so it is a reference row (not input-matched). C-MPJPE is absolute camera-frame, matching our C-MPJPE. ^e All of our numbers come from a single end-to-end feedforward run (our detbox v3, no SLAM, predicted extrinsics, 30-frame WA window); the head is trained jitter-robust, so it tolerates the same noisy detboxes that degrade the offline baselines. Feedforward FPS = 6.59, measured on an NVIDIA GB10 (mean over 30 timed clips: 2427.5 ms per 16-frame forward+lift, the clip length the eval that produces our C/WA/W numbers actually runs; a single 30-frame forward pass is 4.5 fps). Total params 1178.7 M, trainable 262.4 M. ^f Offline FPS is MEASURED, not estimated: the full HaWoR pipeline (detect + DROID-SLAM + Metric3D + motion + infiller) ran 157 sequences of 300-frame video in 5 h 18 min wall-clock on a datacenter GPU = 2.47 native fps, dominated by offline DROID-SLAM (global bundle adjustment). The per-frame hand regressor alone is 6.9 fps (113.4 min for 157 seqs $\times$ 300 frames); SLAM is the bottleneck the three offline rows share. The load-bearing distinction is *online/feedforward* (single causal pass, no future frames, no global optimization) vs *offline* (whole-video SLAM), not raw throughput on matched hardware. ^g HaMeR_{mt}/HaMeR_{ft}: HaMeR fine-tuned on HOI4D with our absolute-depth (kp3d.abs) recipe (mt = metric-tuned translation head; ft = our stronger fine-tune), per-frame on detbox v3, composed with the same DROID-SLAM trajectory. These show our recipe transfers to a HaMeR backbone (fine-tuned HaMeR_{ft} + SLAM reaches C-MPJPE 37.4, on par with our feedforward 36.2) - but they remain *offline* (need whole-video SLAM), whereas ours is a single online pass. Composed SE(3) residual 0.003 mm. ^h Dyn-HaMR input-matched: its per-frame hand estimator seeded from our detbox v3 (WiLoR init) and its DROID-SLAM stage replaced by our SLAM cameras, so the same boxes + trajectory feed it as every other row; PHALP tracking and DROID are bypassed. Dyn-HaMR is an *optimization* method (per-sequence test-time fitting with a motion prior); on ego HOI4D it is unstable - the optimizer drives many sequences to a degenerate world-scale / hand-shape (inflated MANO betas, occasional negative scale), so its *absolute* C-MPJPE and global W-MPJPE blow up (972 / 139 mm) and are omitted as unreliable; only the alignment-invariant WA-MPJPE (48.3) is meaningful. This is not Dyn-HaMR's native regime (it is built for its own PHALP + DROID pipeline). Current subset: 27 sequences (25 converged; extending to 157). Runs on the same offline SLAM trajectory (FPS 2.47^f).